PlantPulse — Incident Report

PlantPulse Demo Site — Process Unit 7 · Generated 2026-07-22 18:52

Incident	Overpressure Risk While the Relief Path Is Degraded
Zone	Z1
Window	2026-07-22T16:00:00 → 2026-07-22T18:00:00
Ground-truth severity	CRITICAL
Single-channel alarm would fire?	No — this is exactly the compound pattern PlantPulse exists to catch

Compound Risk at Peak

83 / 100

Confidence: **47%**

Contributing Factors

- Sensor: +55%
- Permit: +25%
- Maintenance: +10%
- Equipment: +6%
- Weather: +4%

Root Cause (Knowledge Graph)

TANK-01 shows Pressure rising, Tank Temperature rising; VALVE-01 shows Relief Capacity falling; MT-02 (Relief valve function test) is in progress; a hot-work permit is active in the same restricted zone.

SOP Response — SOP-HOTWORK-01: Hot Work Permit Suspension (Ignition Risk During Process Excursion)

Source: OISD-STD-105 (Work Permit System) — hot work permit provisions; Factories Act 1948, Section 38 (Precautions in case of fire)

1. Immediately suspend the active hot-work permit and stop all spark- or flame-producing work
2. Remove hot-work equipment and other ignition sources from the zone
3. Verify fire watch personnel and extinguishing equipment are stationed before any resumption is considered
4. Do not reissue the permit until the pressure/temperature excursion is confirmed resolved and relief capacity is restored to 100%
5. Re-verify the gas-free / LEL certificate before hot work resumes
6. Document the near-miss and route the permit through the issuing authority for re-approval

Estimated risk reduction if followed: **7%** (peak compound risk 83 → 77, computed by re-scoring a mitigated counterfactual through the same fusion model)

Detection Performance

Across 20 independently seeded evaluation days, PlantPulse detected this archetype in **100%** of instances with a median lead time of **38 minutes** over an extrapolated conventional single-sensor alarm. Source: engine/metrics/results.json.